# A3 and LR(0) - Results

**Attempt 1 of 1**

Written Feb 26, 2026 10:23 AM - Feb 26, 2026 11:21 AM

| | |
|---|---|
| Attempt Score | **12 / 12 - 100 %** |
| Overall Grade (Highest Attempt) | **12 / 12 - 100 %** |

## Question 1

1 / 1 point

In A3, In what order must you run the build commands?

- ○ javac → java JLex.Main → java java_cup.Main
- ○ java java_cup.Main → java JLex.Main → javac
- ✓ ◉ java JLex.Main → java java_cup.Main → javac
- ○ javac → java A3User → java JLex.Main

## Question 2

1 / 1 point

in A3, Why can't you compile calc.lex.java immediately after running JLex?

- ○ JLex has a bug that produces broken Java
- ✓ ◉ calc.lex.java references CalcSymbol, which hasn't been generated yet

- ○ You need to rename it to A3Scanner first
- ○ The JDK version is incompatible

**Question 3** 1 / 1 point

Is BEGIN END a valid Block in the TINY grammar?

- ○ Yes, empty blocks are always allowed
- ✓ ◉ No, because

  Block → BEGIN Statement+ END

  requires at least one statement
- ○ Yes, if it's inside a method
- ○ No, because BEGIN must be followed by a semicolon

**Question 4** 1 / 1 point

Which identifier definition is correct for TINY?

- ○ [a-zA-Z_][a-zA-Z0-9_]* — letters, digits, underscores
- ✓ ◉ [a-zA-Z][a-zA-Z0-9]* — letters and digits only, no underscore
- ○ [0-9][a-zA-Z]* — starts with a digit

- [ ] [a-z][a-zA-Z0-9_]* — lowercase start only

**Question 5** 1 / 1 point

When renaming calc to A3, which of the following changes is NOT required?

- [ ] Renaming the file calc.lex to A3.lex
- [ ] Changing CalcScanner to A3Scanner inside the lex file
- [ ] Changing CalcSymbol to A3Symbol inside the lex file
- [x] Changing the JDK installation folder name

**Question 6** 1 / 1 point

What should happen when the scanner encounters an unknown symbol like @?

- [ ] Silently discard it with {}
- [ ] Print an error message to stdout and continue
- [x] Return new Symbol(A3Symbol.error) so the parser rejects the program
- [ ] Throw a Java exception immediately

**Question 7** 1 / 1 point

Why do we need to  remove the RESULT assignment in most rules of A3.cup ?

- ○ It makes the parser faster by skipping evaluation
- ✓ ◉ It changes the parser from evaluating expressions to only recognizing syntax
- ○ It removes the need for terminal declarations
- ○ It prevents the parser from building a parse tree

**Question 8** 1 / 1 point

What is wrong with this rule in A3.cup?

assignStmt ::= ID ASSIGN expr

- ○ ID should be lowercase
- ○ ASSIGN is not a valid terminal name
- ✓ ◉ The rule is missing a semicolon at the end
- ○ expr must be wrapped in parentheses

**Question 9** 1 / 1 point

LR parsers are classified as

- ○ Top-down parsers using leftmost derivation
- ✓ ◉ Bottom-up parsers using rightmost derivation in reverse

- ○ Top-down parsers using rightmost derivation
- ○ Bottom-up parsers using leftmost derivation

**Question 10** 1 / 1 point

The stack of an LR parser stores

- ○ Only terminals
- ○ Only nonterminals
- ✓ ◉ States (and usually grammar symbols interleaved)
- ○ Only input symbols

**Question 11** 1 / 1 point

In an LR parsing table, the ACTION part is indexed by

- ○ (state, nonterminal)
- ✓ ◉ (state, terminal)
- ○ (nonterminal, terminal)
- ○ (terminal, state)

**Question 12** 1 / 1 point

In an LR parsing table, the GOTO part is indexed by

- ( ) (state, terminal)
- (✓) (state, nonterminal)
- ( ) (nonterminal, terminal)
- ( ) (nonterminal, state)

Done